Khalid Zobaid Adnan

Ph.D. candidate, Mechanical Engineering, University of Utah

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github.com/kzadnan · LinkedIn · Advisor: Tianli Feng · Expected completion: summer 2027

RESEARCH FOCUS

I study how heat crosses solid–solid interfaces in wide- and ultrawide-bandgap electronics. The question I am trying to settle is whether thermal boundary conductance is a property of one junction, or of the stack around it. I train machine-learning interatomic potentials on density-functional forces and use them to run nonequilibrium molecular dynamics at lengths direct DFT cannot reach. Four first-author papers in *Physical Review B*, *Physical Review Applied*, *Applied Physics Letters*, and the *International Journal of Heat and Mass Transfer*.

SELECTED RESULTS

- A nearby interface is not a spectator. In Si/Ge, adding one raised thermal boundary conductance from 400 to 700 MW m^{-2} K^{-1}. A 30 nm Si film between Ge increased apparent thermal conductivity from 50 to 280 W m^{-1} K^{-1}, above bulk Si, by filtering which phonon modes carry the heat (*Phys. Rev. B* 2024).
- The same idea holds in GaN/AlN: a standalone interface is about 600 MW m^{-2} K^{-1} after quantum correction; a second nearby interface can raise it to about 1000 by ballistic filtering, not by forming superlattice minibands (*Phys. Rev. B* 2025).
- A DFT-trained potential was required to match Si/diamond experiment (130 vs. ~140 MW m^{-2} K^{-1}; Tersoff molecular dynamics gave 310). A 1 nm SiN_x interlayer raised conductance to 275, and amorphous carbon to 350. In a silicon-on-diamond device model, lifting conductance from 50 to 350 cut the hot-spot temperature rise by tens of kelvin (*Phys. Rev. Applied* 2026).
- An AlGaN interlayer between GaN and AlN lowers conductance across the alloy range; alloy scattering beats the usual phonon-bridge picture. A sigmoidal composition grade recovers it (*Appl. Phys. Lett.* 2026).
- With U.S. Army DEVCOM, quantum-corrected metal/diamond conductances for Al, Mo, Zr, and Au are about 284, 93, 30, and 40 MW m^{-2} K^{-1}. They track the metal Debye temperature more than elastic modulus (*Int. J. Heat Mass Transfer* 2024).

EDUCATION

Ph.D., Mechanical Engineering — University of Utah 2023 – present
Thesis: interfacial thermal transport and the interplay between successive interfaces in solid–solid heterostructures. Advisor: Tianli Feng.

M.S., Mechanical Engineering — University of Utah 2023 – 2025
GPA 4.0/4.0. Summa Cum Laude (top 1% of graduates in a five-year window).

B.Sc., Mechanical Engineering — Bangladesh University of Engineering and Technology 2016 – 2021
CGPA 3.53/4.00.

RESEARCH EXPERIENCE

Graduate Research Assistant — Feng Research Group, University of Utah 2023 – present

- Own the successive-interface calculations: design the cells, choose the spacer against the phonon mean free path, and separate real interface–interface interplay from finite-size artifacts. That protocol is what the later GaN/AlN, Si/diamond, and GaN/diamond studies are built on.
- Train DeePMD and moment-tensor potentials on DFT forces for chemically abrupt, alloyed, and metal–diamond interfaces, then drive nonequilibrium molecular dynamics in cells direct first-principles dynamics cannot reach. DFT reference cells are typically 300–500 atoms.
- Put classical molecular dynamics, the diffuse mismatch model, and experiment on one scale by implementing a quantum correction for mode occupation. Without it, room-temperature comparisons for stiff crystals such as diamond are not meaningful.
- Carry atomistic conductance into a finite-element silicon-on-diamond device model so a change in interfacial resistance shows up as hot-spot temperature, not only as a materials number.
- Collaborate with U.S. Army DEVCOM Army Research Laboratory on metal/diamond contacts (published) and with Sandia National Laboratories on GaN/diamond thermal boundary conductance (in progress).

Supported by NSF CAREER award CBET-2337749, “Prediction and understanding of thermal transport across successive interfaces” (PI: Tianli Feng, 2024–2029). Computing: Center for High Performance Computing, University of Utah, and Bridges-2 at the Pittsburgh Supercomputing Center through NSF ACCESS.

PUBLICATIONS

1. **Khalid Zobaid Adnan**, H. Zhou, and T. Feng, “Significant impact of $Al_{1-x}Ga_xN$ interlayer on GaN/AlN thermal boundary conductance,” *Appl. Phys. Lett.* **128**, 122201 (2026). 10.1063/5.0323350
2. **Khalid Zobaid Adnan**, T. Abedien, and T. Feng, “Si/diamond thermal boundary conductance enhanced by SiN_x and amorphous carbon interlayers,” *Phys. Rev. Applied* **25**, 044013 (2026). 10.1103/84jf-svtg
3. H. Zhou, **Khalid Zobaid Adnan**, W. A. Jones, and T. Feng, “Thermal boundary conductance in standalone and non-standalone GaN/AlN heterostructures predicted using machine-learning interatomic potentials,” *Phys. Rev. B* **112**, 235308 (2025). 10.1103/w7qp-tl6z
4. **Khalid Zobaid Adnan**, M. R. Neupane, and T. Feng, “Thermal boundary conductance of metal–diamond interfaces predicted by machine learning interatomic potentials,” *Int. J. Heat Mass Transfer* **235**, 126227 (2024). 10.1016/j.ijheatmasstransfer.2024.126227
5. **Khalid Zobaid Adnan** and T. Feng, “Thermal boundary conductance and thermal conductivity strongly depend on nearby environment,” *Phys. Rev. B* **109**, 245302 (2024). 10.1103/PhysRevB.109.245302

PRESENTATIONS

- “Thermal boundary conductance of metal–diamond interfaces predicted by machine learning interatomic potentials,” ASME Summer Heat Transfer Conference, 2025. Conference paper and oral presentation.
- “Thermal boundary conductance and thermal conductivity strongly depend on nearby environment,” ASME Summer Heat Transfer Conference, 2025. Poster.

OPEN RESEARCH SOFTWARE

- MLIPs for interfaced materials — DeePMD and moment-tensor potentials for heterogeneous interfaces, including Si/diamond and Au/diamond, packaged for LAMMPS.
- JAX-MLIP — harmonic-constrained potentials that learn force-constant structure from displacements.
- Autoencoder DMD — mode extraction and reconstruction of ab initio molecular-dynamics trajectories.

TEACHING

Graduate teaching assistant — Mechanical Engineering, University of Utah F23, F25, S26
Numerical Methods; Programming for Modeling and Simulation; Thermodynamics. Lab sections of about 60 students: homework sessions, office hours, project and exam grading.

Lecturer — Daffodil International University, Dhaka 2021 – 2022
Wrote the first outcome-based curriculum draft for the introductory mechanical and mechatronics sequence.

TECHNICAL SKILLS

Atomistics	VASP, Quantum ESPRESSO, LAMMPS, GPUMD, Phonopy, DeePMD-kit, MTP, FitSNAP
Computing	Python, C/C++, MATLAB, JAX, PyTorch, SLURM, multi-GPU workflows
Device scale	COMSOL, ANSYS; finite-element coupling of interfacial resistance to chip temperature